Powerset

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The problem

- Every subset of a set of distinct ints (iteratively doubling).

Intuition

- Each new element doubles the subsets: every existing subset with or without it.

Approach

- 1. Start with the empty set
- 2. For each element, copy all subsets and add it
- 3. Return all

Java solution

```java
public class Powerset {
    public static List<List<Integer>> powerset(List<Integer> array) {
        List<List<Integer>> subsets = new ArrayList<>();
        subsets.add(new ArrayList<>());
        for (int x : array) {
            int size = subsets.size();
            for (int i = 0; i < size; i++) { List<Integer> s = new ArrayList<>(subsets.get(i)); s.add(x); subsets.add(s); }
        }
        return subsets;
    }
}
```

Complexity

- Time: O(n * 2^n)
- Space: O(n * 2^n)

Recap

- Each new element doubles the subsets: every existing subset with or without it.
- Complexity: O(n * 2^n)